

A close-up photograph of an internal combustion engine's piston and crankshaft. The piston is at the top, showing its rings and crown. The crankshaft is below it, with a visible connecting rod. The background is a blurred blue and white, suggesting a workshop or engine bay environment.

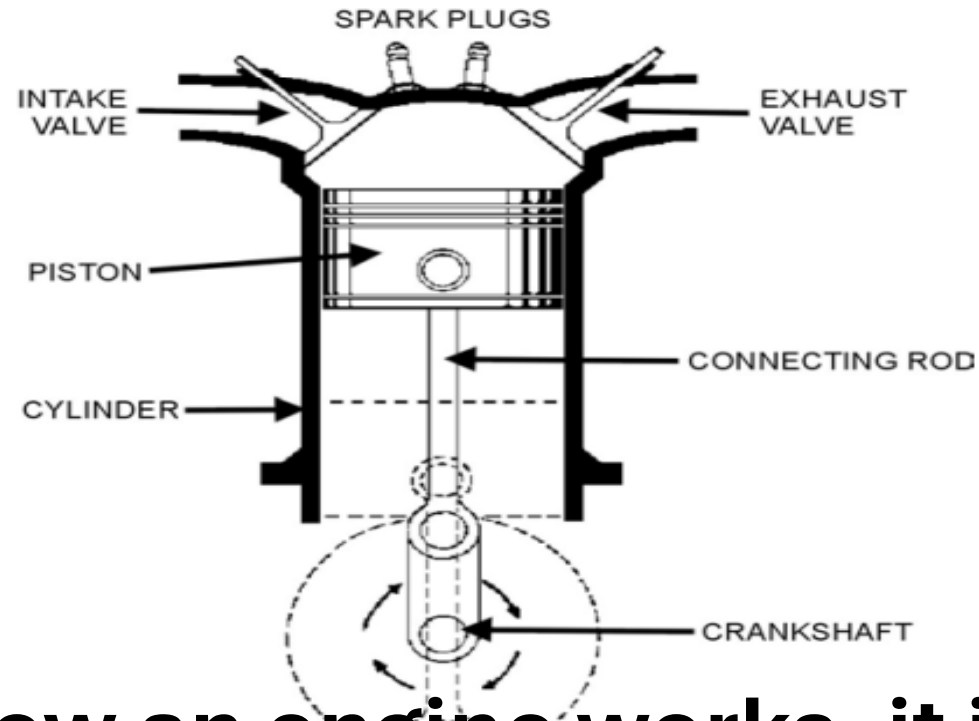
EO M232.02 – IDENTIFY THE
COMPONENTS OF
PISTONPOWERED INTERNAL
COMBUSTION ENGINES

Teaching Points

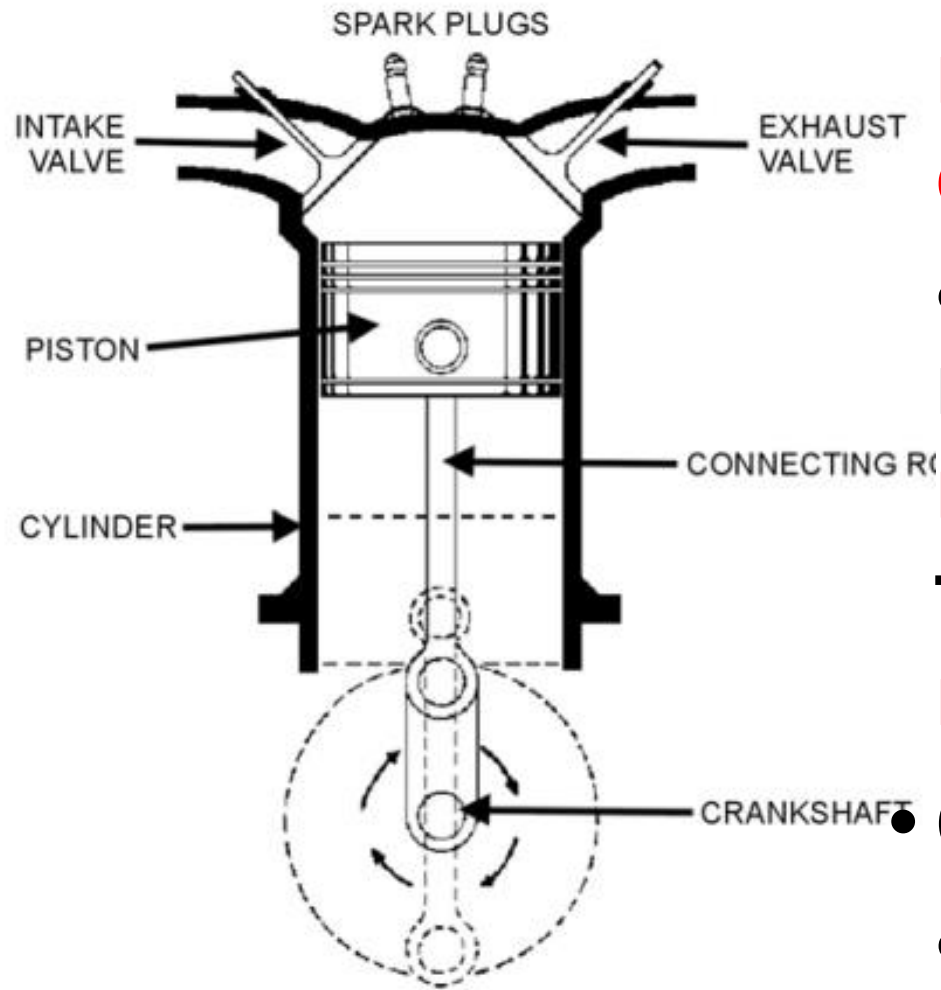
- TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine
- TP2 Explain the Difference Between Rotary Engines and Radial Engines
- TP3 Describe Aircraft Engine Arrangements
- TP4 Play a Game of Piston-powered Baseball



TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine pt1



- **Cylinder:** To understand how an engine works, it is necessary to first know what parts make up an engine. The cylinder is the **main component**. This is where the **combustion of a gasoline and air mixture** takes place
- **Piston:** The piston is found in the cylinder and is **driven up and down** by the exploding air and fuel mixture

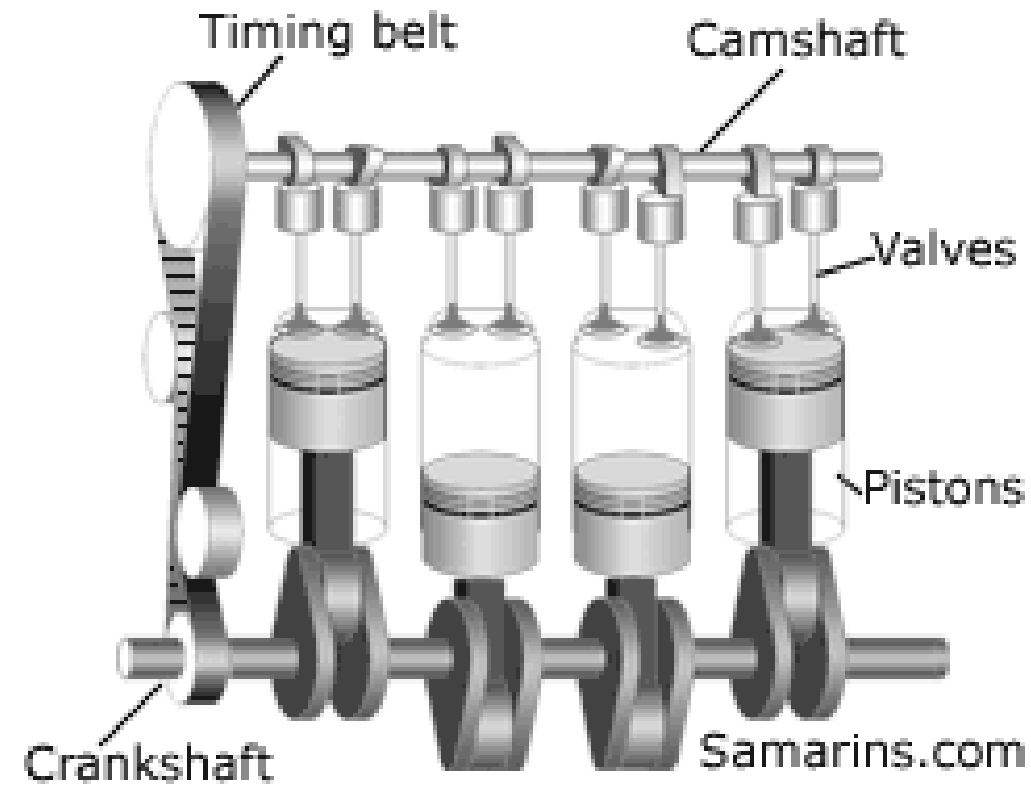


TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine pt2

- **Connecting Rod:** The connecting rod is joined to the piston and to the crankshaft with bearings which allow movement so that the reciprocating (up-and-down) motion of the piston can be transformed into rotary (spinning) motion of the crankshaft
- **Crankshaft:** As the piston drives up and down, the connecting rod rotates around the crankshaft, turning it. The crankshaft can rotate while the piston goes up and down

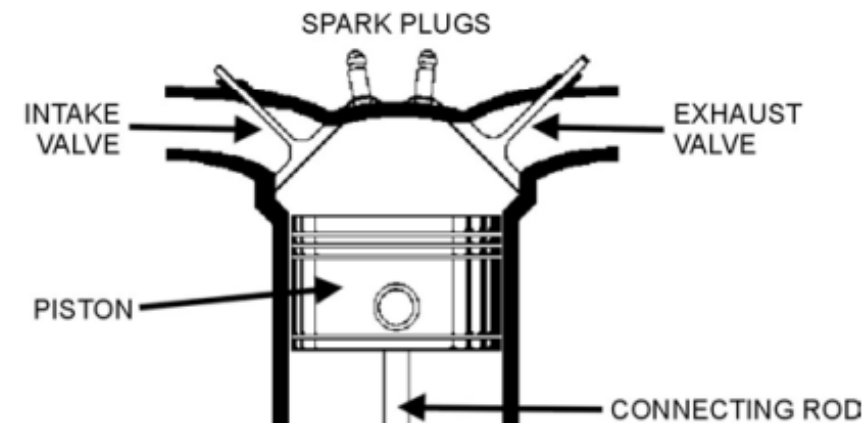
TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine

pt3



- **Camshaft:** The crankshaft often turns a **second shaft called a camshaft**
- The **cams** are bumps on the camshaft that **open and close the intake and exhaust valves** at the correct time
- Each cylinder has at least one **set of valves operated by the cams** on the camshaft
- **Intake valve:** opens to let the **mixture of gasoline and air into the cylinder** and then closes.
- **Exhaust valve:** **opens to release the exhaust** after mixture is burnt and then closes

TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine pt4

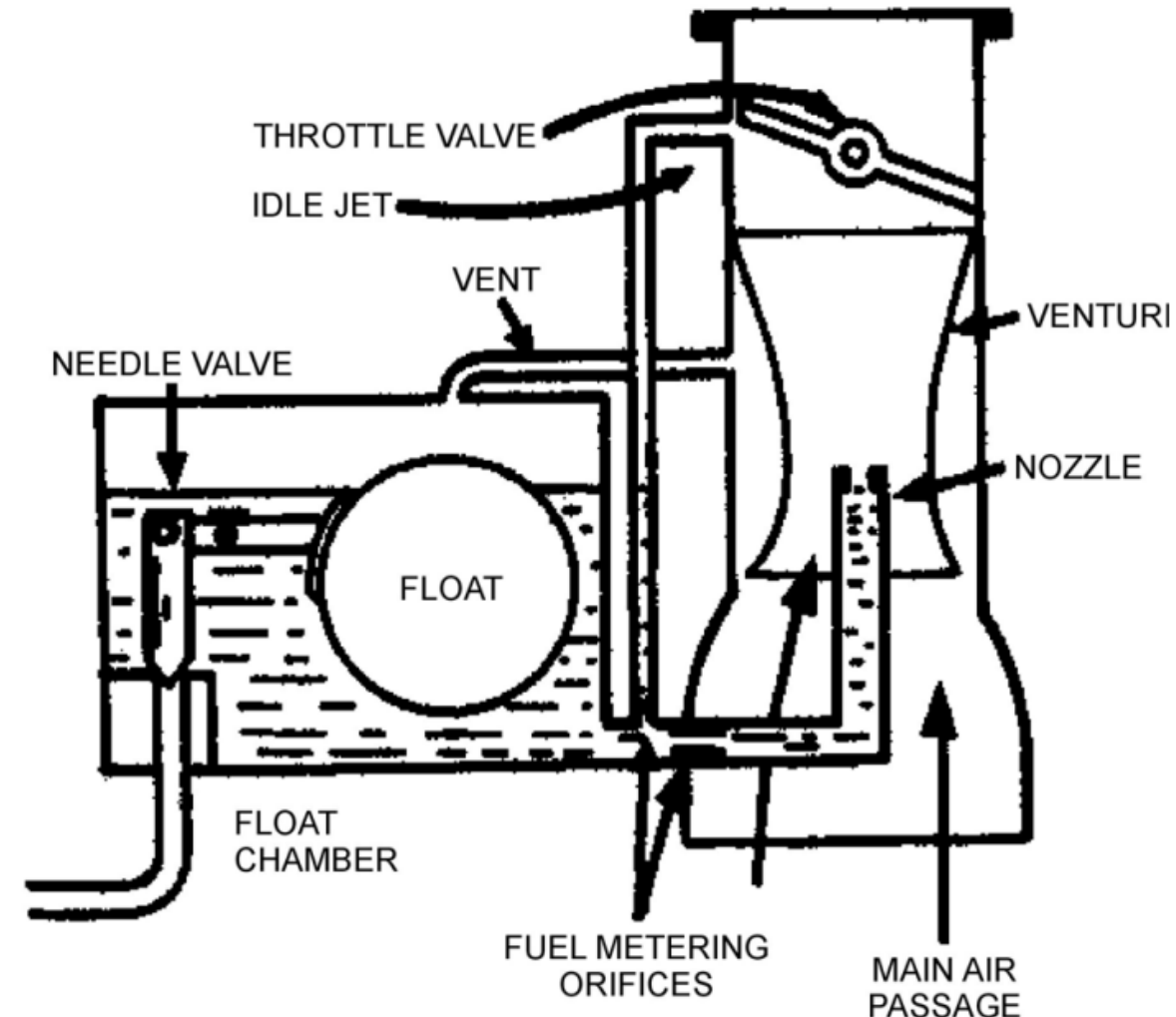


- Distributor: The **gasoline and air mixture is ignited by a spark plug**. Most aircraft have 2 spark plugs in every cylinder
- The **fuel takes time to burn completely**. Because of this, the spark must happen at a **fraction of a second** before the piston has reached the top. In a multi-cylinder engine such as aircraft use, an **electrical signal** must be sent to each cylinder's spark plug at exactly the right time
- The **timing and distribution of spark** sometimes relies on a central distributor, which is worked by gears from the crankshaft. Should this **distributor fail, engine will stop**
- A better, though more expensive method, is to equip each cylinder with its **own spark timing and delivery system**

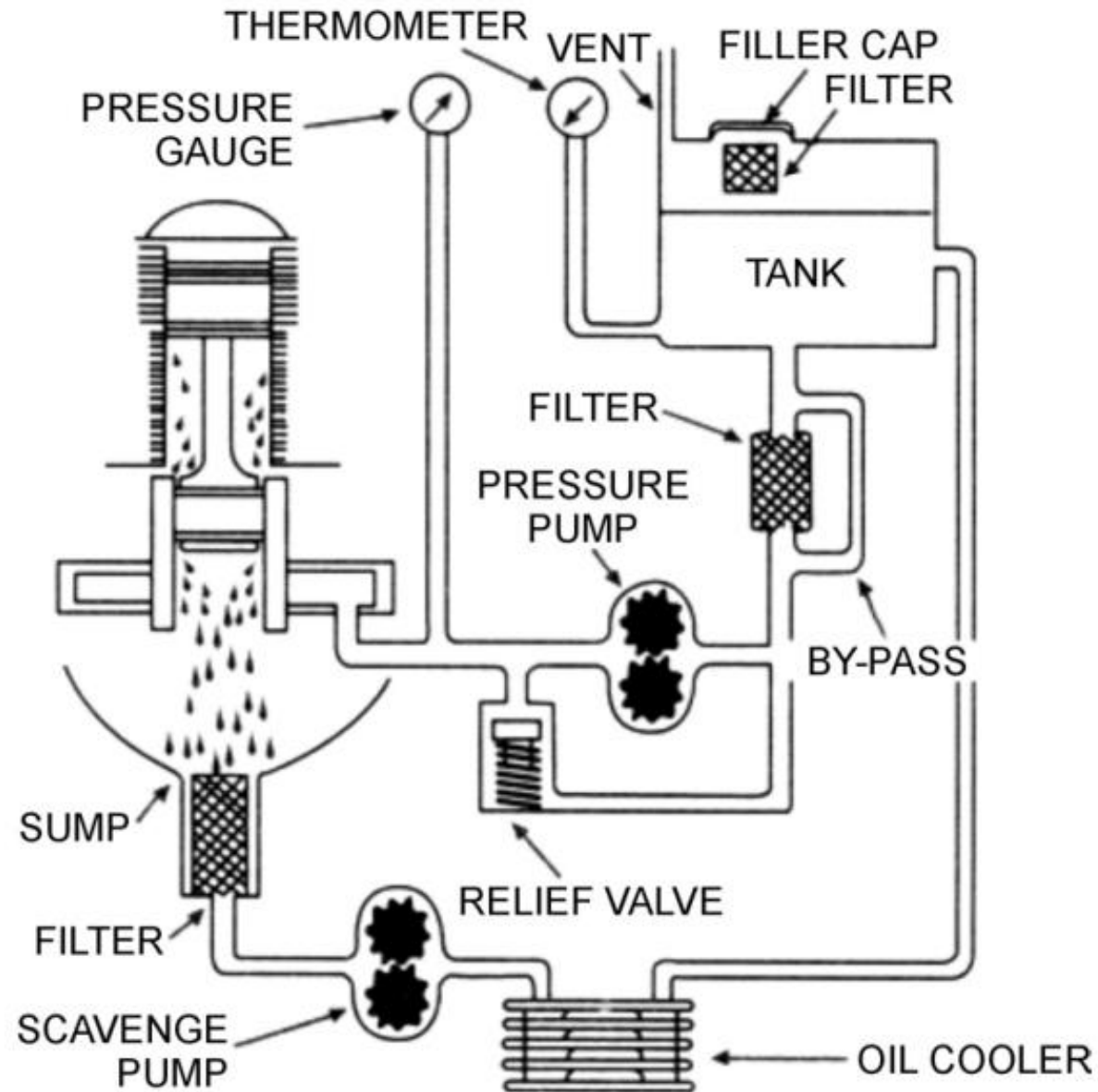
- **Carburetor: Before fuel is delivered to the cylinder for detonation, it is mixed with **air in exact proportion**. A fuel injector or a carburetor does this. For effective detonation and clean burning, the fuel must be **broken into tiny droplets** and mixed with air**

TP1 Identify and Explain the Operations of the Major Components of a Piston-powered Engine pt5

CARBURETOR



DRY-SUMP LUBRICATION SYSTEM



- **Oil Sump.** The moving parts of the engine all need to be **coated with engine oil**. Oil is provided under pressure to make sure that all moving parts are coated. A **wet sump** stores the oil supply in the crankcase with the crankshaft, while a **dry sump** stores the oil in a **separate tank** and delivers it to the engine via piping

Which component of a piston-powered engine transforms the reciprocating motion of the piston into rotary motion?

A. Camshaft

B. Crankshaft

C. Connecting Rod

D. Distributor

The intake valve in a piston-powered engine is responsible for:

A. Igniting the fuel mixture

B. Taking in the exhaust gases

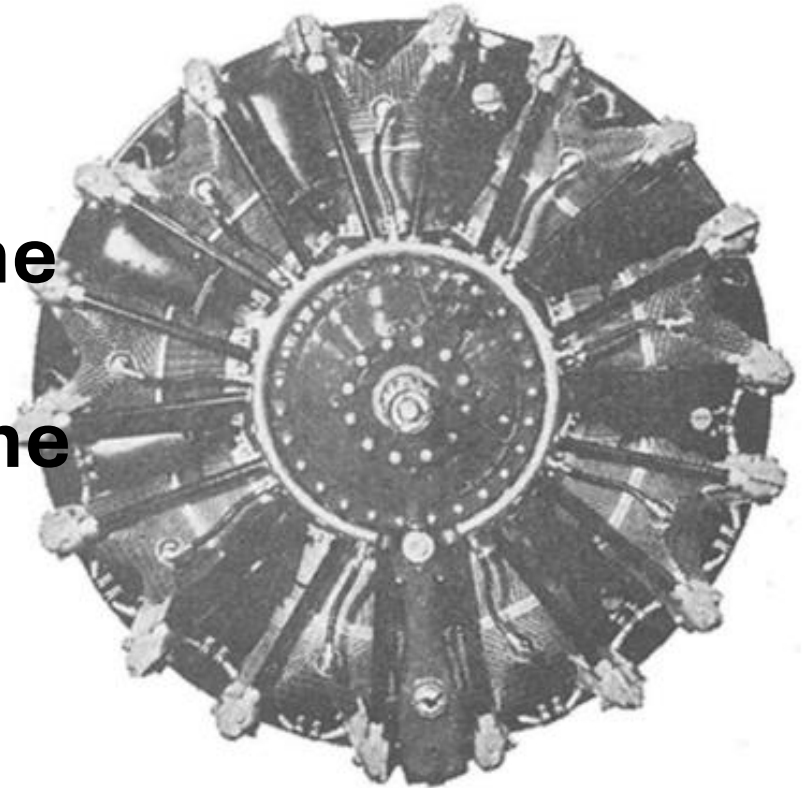
C. Allowing the mixture of gasoline and air into the cylinder

D. Transforming reciprocating motion into rotary motion

TP2 Explain the Difference Between Rotary Engines and Radial Engines pt1

- Some early aircraft engines used rotary engines in which the **cylinders rotated around the stationary central crankshaft**
- These were different than the later radial engines which the **stationary cylinders were arranged around the rotating crankshaft**
- Many larger older aircraft had radial engines. In this design the **cylinders were arranged in a circle** at the front of the engine with the cylinder tops pointed outwards. The **crankshaft ran through the middle** of the cylinders to the front of the aircraft. Radial engines had many cylinders; some aircraft from World War II had 13 cylinders.

RADIAL ENGINE



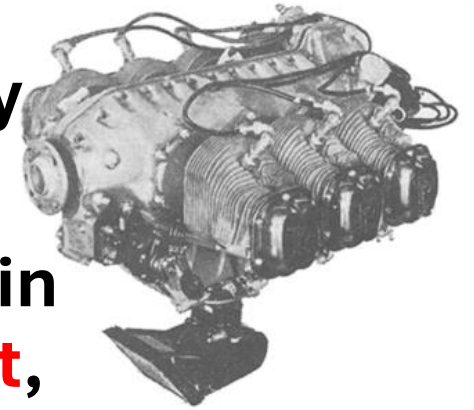
- **Even older aircraft, before and during World War I, had rotary engines that were different but were often confused with the later radial types. In the rotary engine, the crankshaft was stationary and the cylinders rotated around the crankshaft. This is the opposite of the radial engine, which had stationary cylinders and rotating crankshaft. Rotary engine design was abandoned because the great weight of the spinning cylinders was found to interfere with turning the aircraft in flight.**

TP2 Explain the Difference Between Rotary Engines and Radial Engines pt2

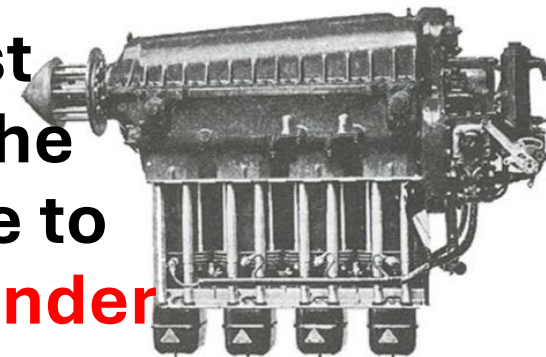
- What is the key difference between rotary engines and radial engines in early aircraft designs?
- A. Rotary engines had stationary cylinders while the crankshaft rotated.
- B. Rotary engines had rotating cylinders around a stationary crankshaft.
- C. Radial engines had rotating cylinders around a stationary crankshaft.
- D. Both engines had rotating crankshafts and cylinders.

TP3 Describe Aircraft Engine Arrangements

- Internal combustion engines come in a **variety of styles** by how the cylinders are configured.
- The horizontally opposed engine is most commonly used in airplanes. This engine has **two banks of cylinders lying flat**, directly opposite each other and **working on the same crankshaft**. There may be 4, 6 or 8 cylinders. The advantage of this engine type is its **flat shape** that generates **less form drag**. Form drag is a force that opposes the aircraft's movement through the air.
- Some older aircraft have in-line engines. This was the first type of aircraft engine used greatly. In an in-line engine, the **cylinders are lined up in a row** from the front of the engine to the back, with the tops pointed up. The **crankshaft runs under the cylinders** to the front of the aircraft.



IN-LINE ENGINE
IN-LINE



Which aircraft engine arrangement is most commonly used in airplanes due to its flat shape and reduced form drag?

A. In-line Engine

B. Rotary Engine

C. Radial Engine

D. Horizontally Opposed Engine

In an in-line engine, how are the cylinders arranged?

A. In a circle with tops pointing outwards

B. Flat and opposite each other

C. Lined up in a row from front to back with tops pointed up

D. Rotating around a stationary crankshaft